

Software Design & Architecture

Semester Project



Instructor

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Submitted by

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Section

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Department of Software Engineering

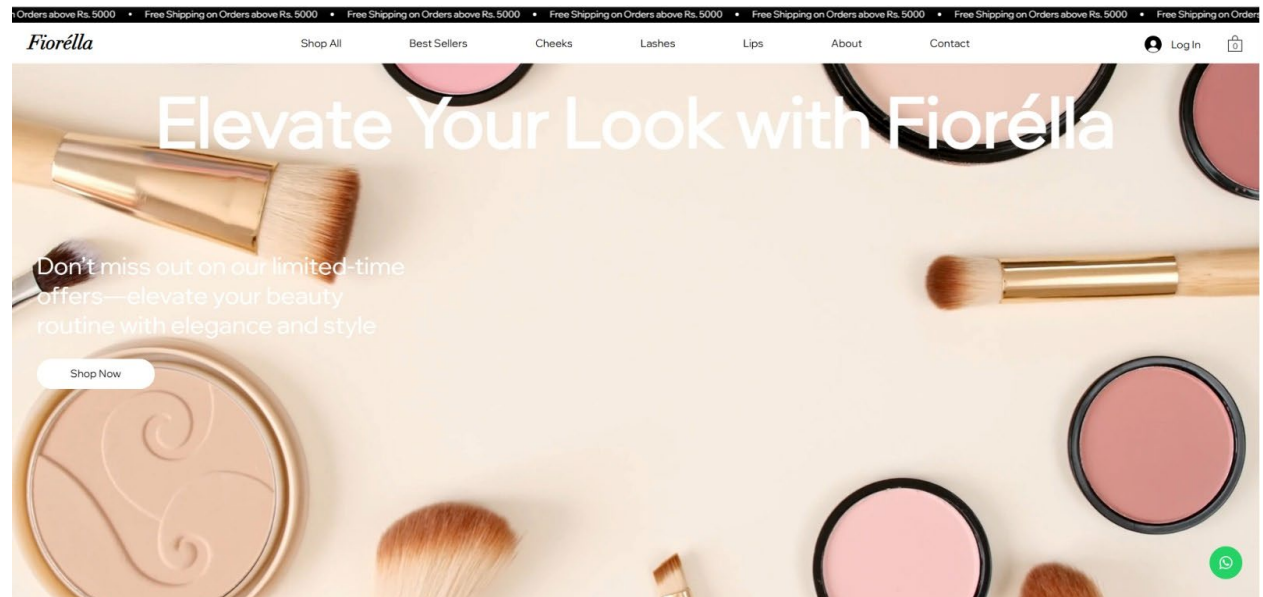
The University of Lahore

Lahore, Pakistan

[25th May, 2025]

Introduction

Fiorella Beauty is an online store made on **Wix Studio**. It sells makeup products. The goal is to create a beautiful, easy-to-use site where people can browse, learn about products, and buy them easily.



BEST SELLERS



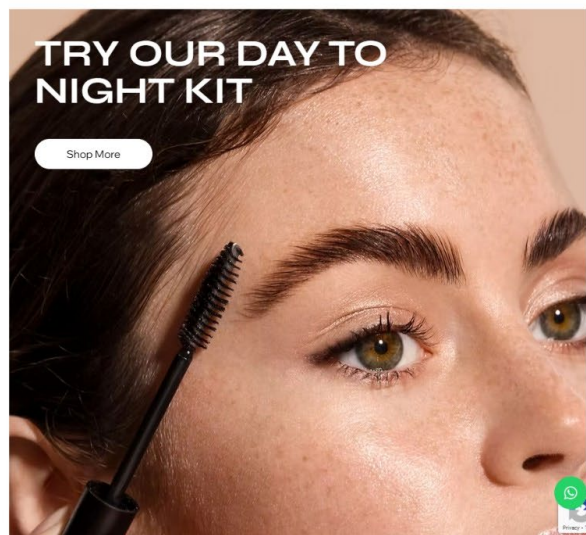
Celeste Glitter
~~Rs.1,500.00~~ Rs.1,350.00
No reviews

Add to Cart



Pomegranate Magic
~~Rs.1,900.00~~ Rs.1,710.00
No reviews

Add to Cart



Problem Statement:

- Customers want to shop for beauty products anytime, anywhere.
- Visiting stores can be hard or time-consuming.
- Businesses need to grow and reach more customers.
- Manual order handling is slow.
- Building a secure site from scratch is difficult.

Solution:

Fiorella Beauty's website solves all of these by using Wix – it's fast, secure, and easy to manage.

Objectives:

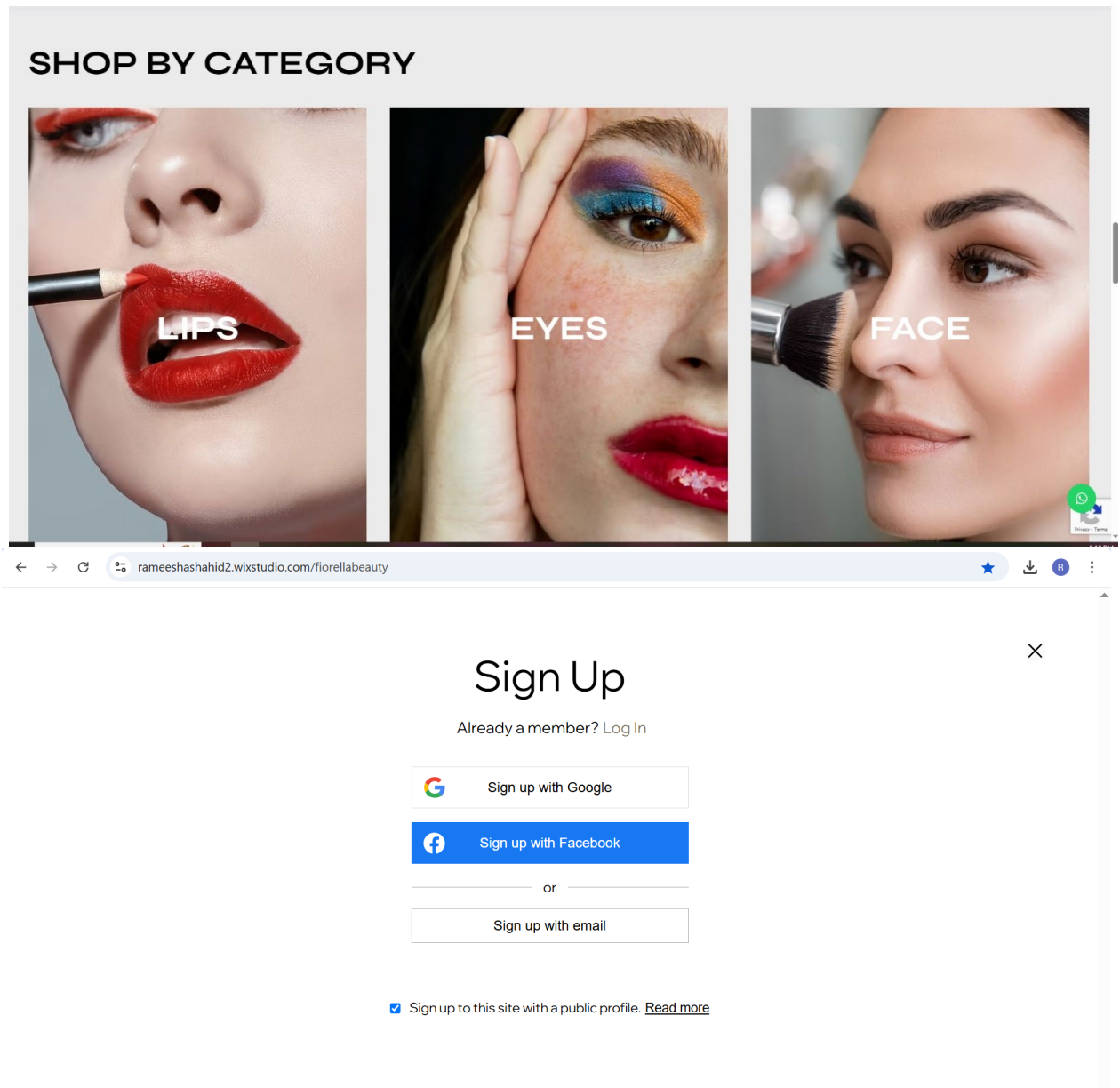
- Fast loading: under 2 seconds
- 99.9% uptime
- Handle 500+ users at once
- Boost sales by 20% in 6 months
- Secure payments (PCI compliant)
- Easily manage 500+ products

Functional Requirements:

These are features the system must provide:

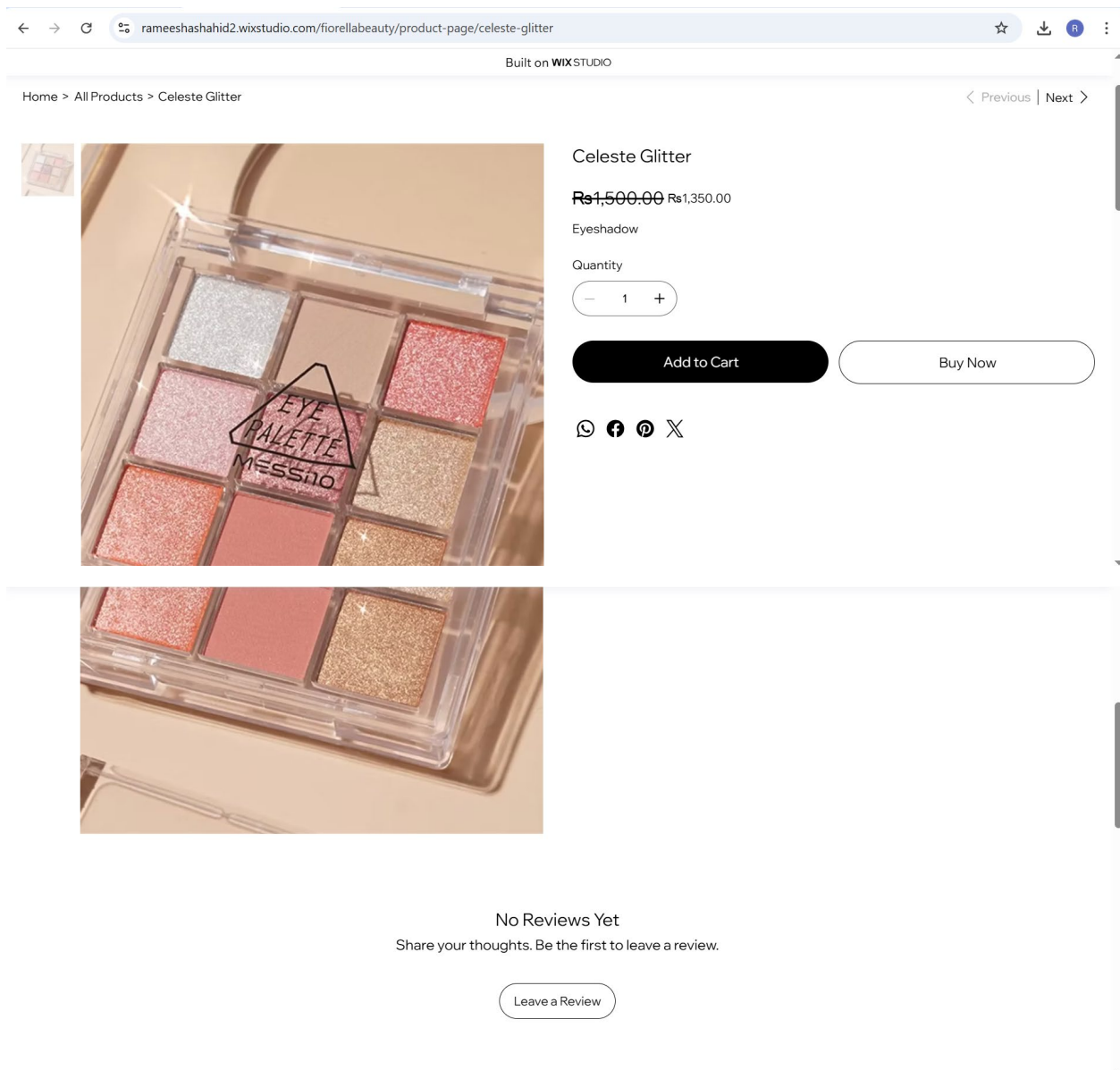
1. User Account Management

- Users can register, login, and logout.
- Users can reset or recover passwords.
- Users can update profile details (name, email, phone number, address).



2. Product Catalog

- Display a categorized list of makeup products (e.g., Lips, Eyes, Face).
- Each product has a detailed page: image, price, description, ratings/reviews.



3. Search and Filter

- Users can search for products by name or category.
- Filters based on price range, product type, bestsellers, etc.

Cart (4 items)

×



Midnight Rose

~~Rs1,900.00~~ Rs1,710.00

— 1 +

Rs1,710.00

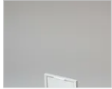


Celeste Glitter

~~Rs1,500.00~~ Rs1,350.00

— 2 +

Rs2,700.00



Blushfire

~~Rs900.00~~ Rs792.00

Subtotal

Rs5,202.00

View Cart

Browse by

16 products

[All Products](#)

Best Sellers

Eyes

Face

Lips

Filter by

Price —

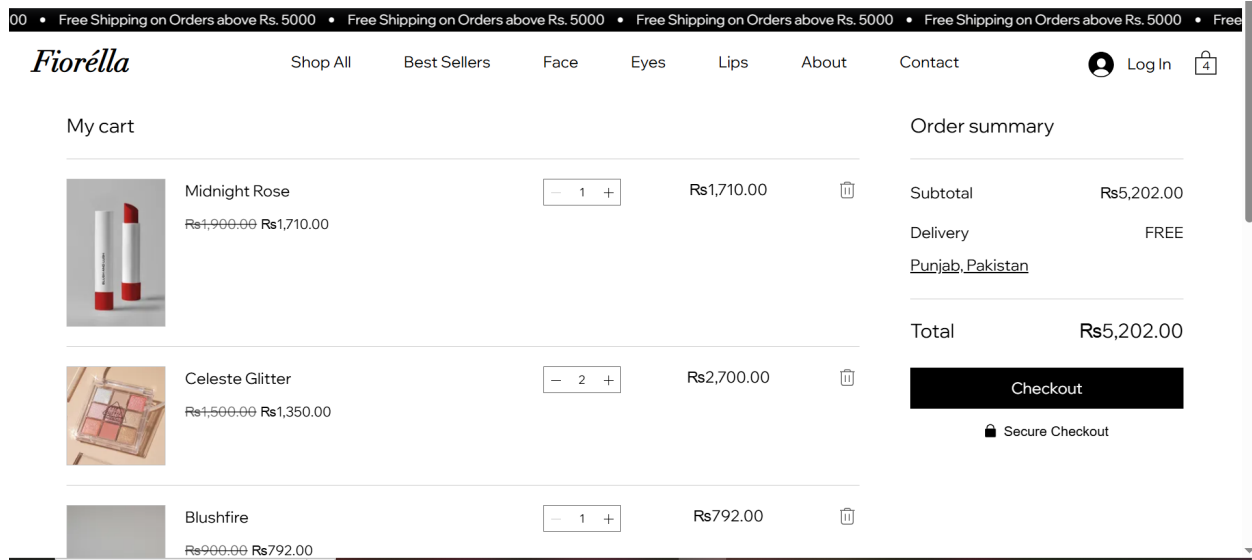
PKR 792 PKR 1,710

4. Shopping Cart

- Users can add/remove products to/from the cart.
- View cart items with total price calculation.
- Update quantities of products in the cart.

5. Checkout Process

- Secure checkout process with address, shipping method, and payment options.
- Order summary before final purchase.



6. Payment Integration

- Integration with payment gateways (e.g., credit/debit card, digital wallets).
- Order confirmation upon successful payment.

7. Order Management

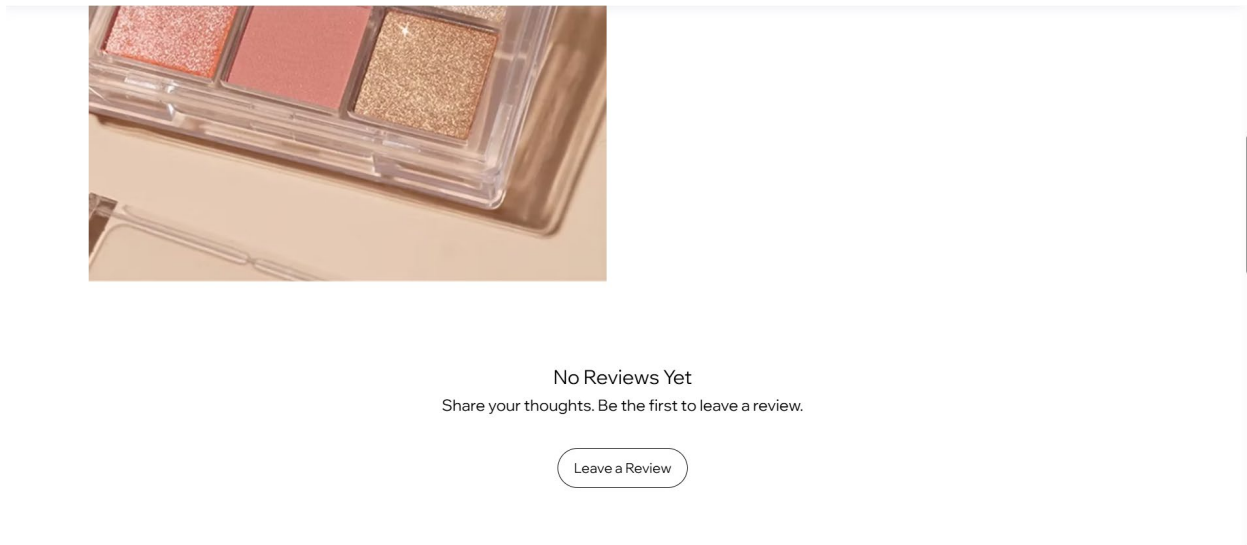
- View past and current orders in the user dashboard.
- Admin can manage all orders: update order status (Processing, Shipped, Delivered).

8. Admin Panel

- Admin login for managing site content.
- Add/edit/delete products.
- View/manage user orders.
- Manage product inventory.

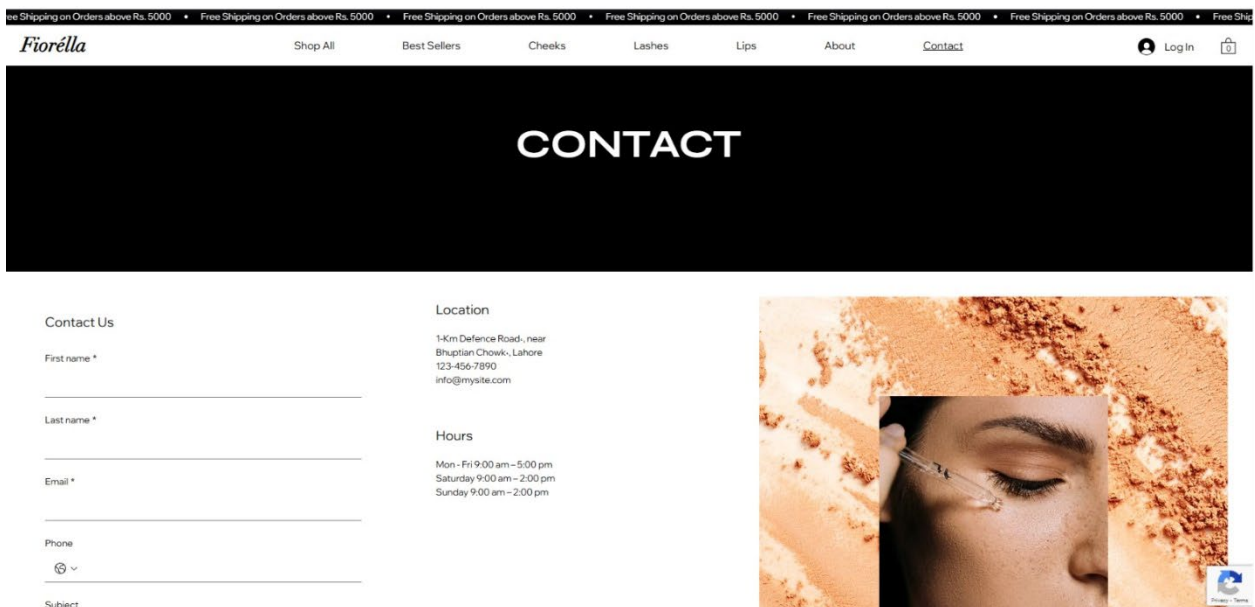
9. Reviews & Ratings

- Users can leave reviews and rate products.
- Display average ratings on product pages.



10. Contact and Support

- Contact form for customer queries.
- Display email and phone support information.



Fiorella

[Shop All](#)

[Best Sellers](#)

[Face](#)

[Eyes](#)

[Lips](#)

[About](#)

[Contact](#)



Log In



Phone



Subject

Message

Submit



Phone



Search

Afghanistan +93

Åland Islands +358

Albania +355

Algeria +213

American Samoa +1684

Fiorella

Menu

[Shop All](#)
[Best Sellers](#)
[Face](#)
[Eyes](#)
[Lips](#)
[About](#)
[Contact](#)

Policies

[FAQ](#)
[Terms & Conditions](#)
[Privacy Policy](#)
[Shipping Policy](#)
[Refund Policy](#)
[Cookie Policy](#)
[Accessibility Statement](#)

Contact

1-Km Defence Road, near
Bhupian Chowk, Lahore

123-456-7890
info@mysite.com

Social

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[Instagram](#)
[Youtube](#)
[X](#)

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We accept the following payment methods



These payment methods are for illustrative purposes only. Please update this section to reflect the payment methods you actually accept, which are



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Non-functional Requirements:

These define how the system performs, their qualities and constraints.

1. Performance

- Pages should load within 2-3 seconds.
- Support up to 1000+ concurrent users during high traffic (e.g., sales).

2. Scalability

- System should support more products, users, and categories as the business grows.

3. Security

- SSL encryption for secure browsing and transactions.
- Passwords must be encrypted.
- Protection against SQL injection, XSS, and CSRF attacks.

4. Availability

- 99.9% uptime, especially during promotional campaigns.
- Regular backup of site data to prevent data loss.

5. Usability

- Intuitive user interface with easy navigation.
- Mobile responsive design.
- Consistent look and feel across all devices.

6. Maintainability

- Code should be modular and well-documented.
- Regular updates for new features and bug fixes.

7. Compatibility

- Support for major browsers (Chrome, Firefox, Safari, Edge).
- Mobile and tablet compatibility.

8. Localization (Optional)

Support for multiple languages and currencies if expanding internationally.

1. **Wix Studio (with Velo):**

Drag-and-drop builder with optional custom coding for enhanced design and functionality.

2. **JavaScript (ES2020):**

Used for any custom interactive features or dynamic content.

3. **HTML5 & CSS3:**

Ensures responsive and visually appealing layouts across all devices.

4. **Wix CMS (Data Collections):**

Manages structured content like product listings and categories, enabling dynamic page updates.

5. **Payment Gateways:**

Supports secure transactions through platforms like PayPal, Visa, MasterCard, and others.

6. **Wix Store:**

Provides shopping cart, checkout, and order management functionalities.

7. **Hosting & Infrastructure:**

Wix handles hosting with a global CDN, ensuring fast load times and reliable performance.

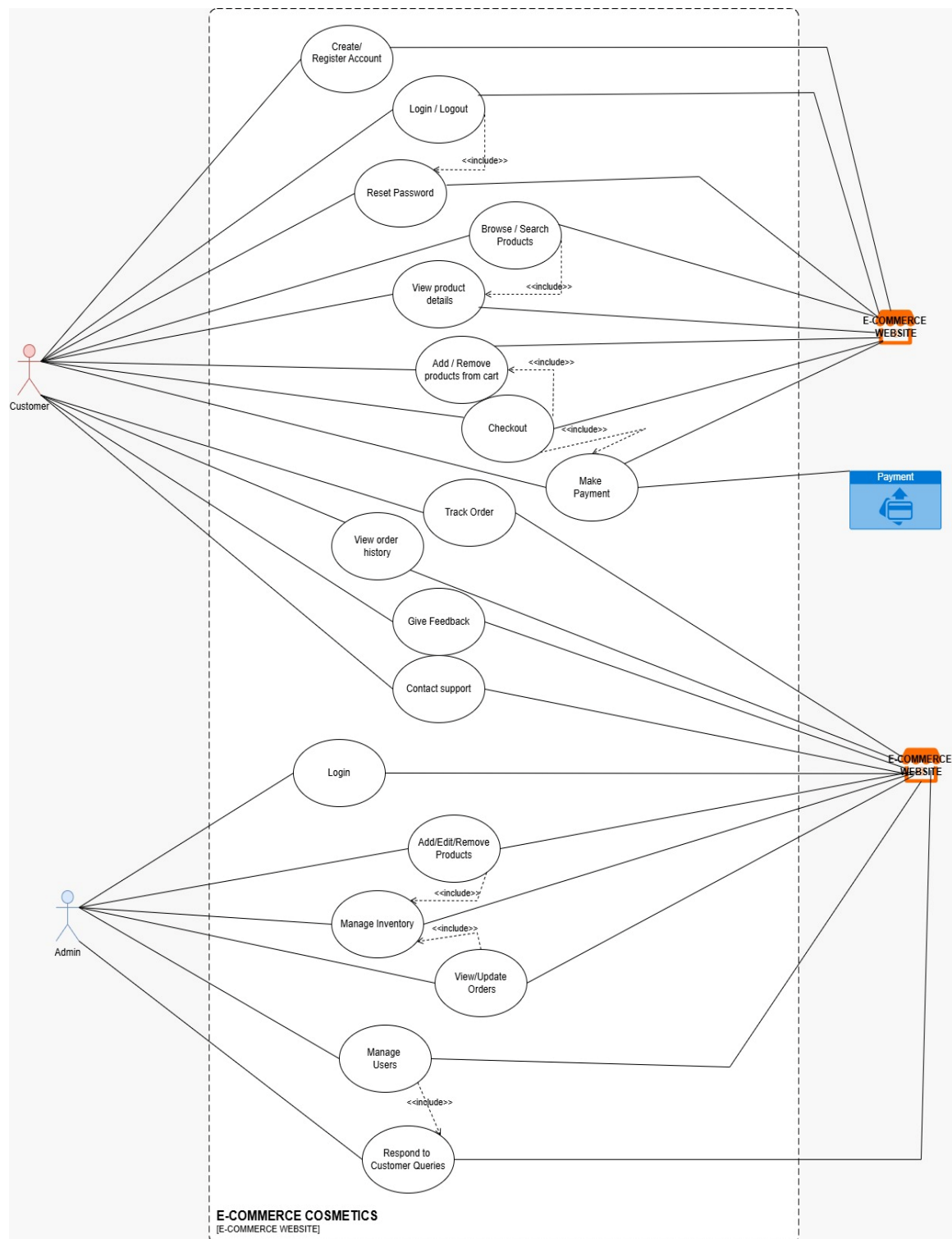
Discussion

The Fiorella Beauty website is a scalable, cloud-based e-commerce solution built with **Wix Studio**. It uses **dynamic pages**, **CMS collections**, and a **serverless backend** to provide a clean separation between business logic and presentation. It ensures a smooth user experience through responsive design, secure payments, and fast performance—all without requiring manual server management.

System Modeling

1. Use Case Diagram

The Use Case Diagram outlines the system's key functionalities, such as user registration, product browsing, shopping cart management, and order processing. It captures the interactions between users (customers and admins) and the system.



2. Data Flow Diagrams (DFDs)

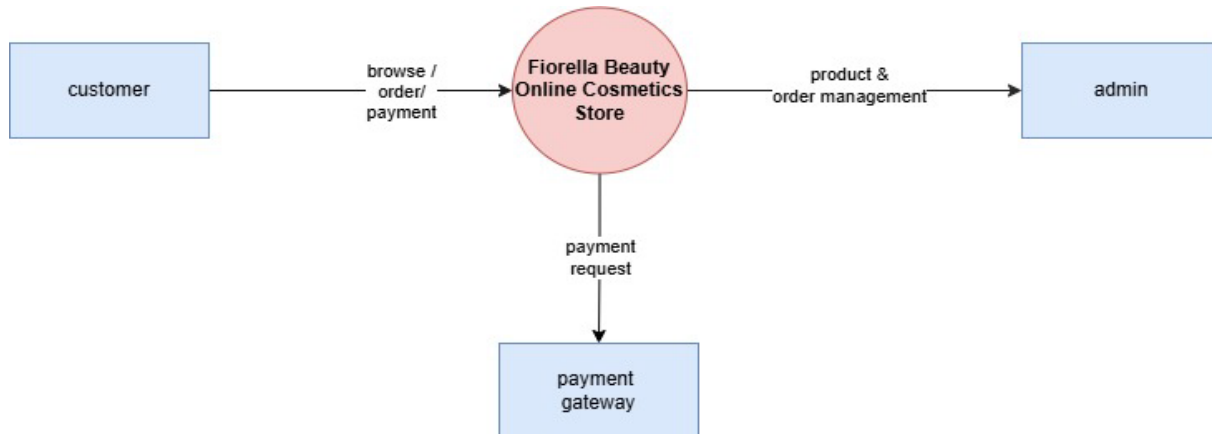
DFDs show how data moves through the Fiorella Beauty website.

They map user interactions, backend processes, and data flows, helping find and fix problems for a smooth user experience.

- **Level 0 Diagram (Context Diagram):**

Shows the whole website as one process.

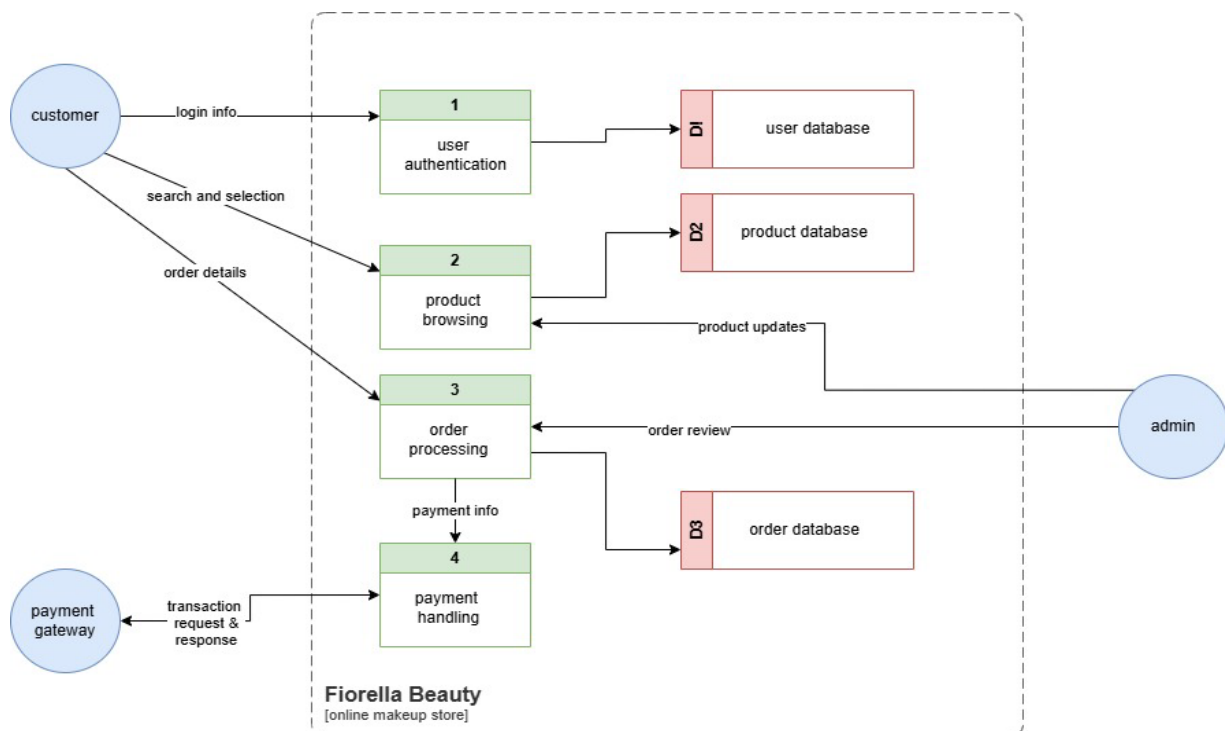
It highlights interactions between users, payment gateways, and the system.



- **Level 1 Diagram:**

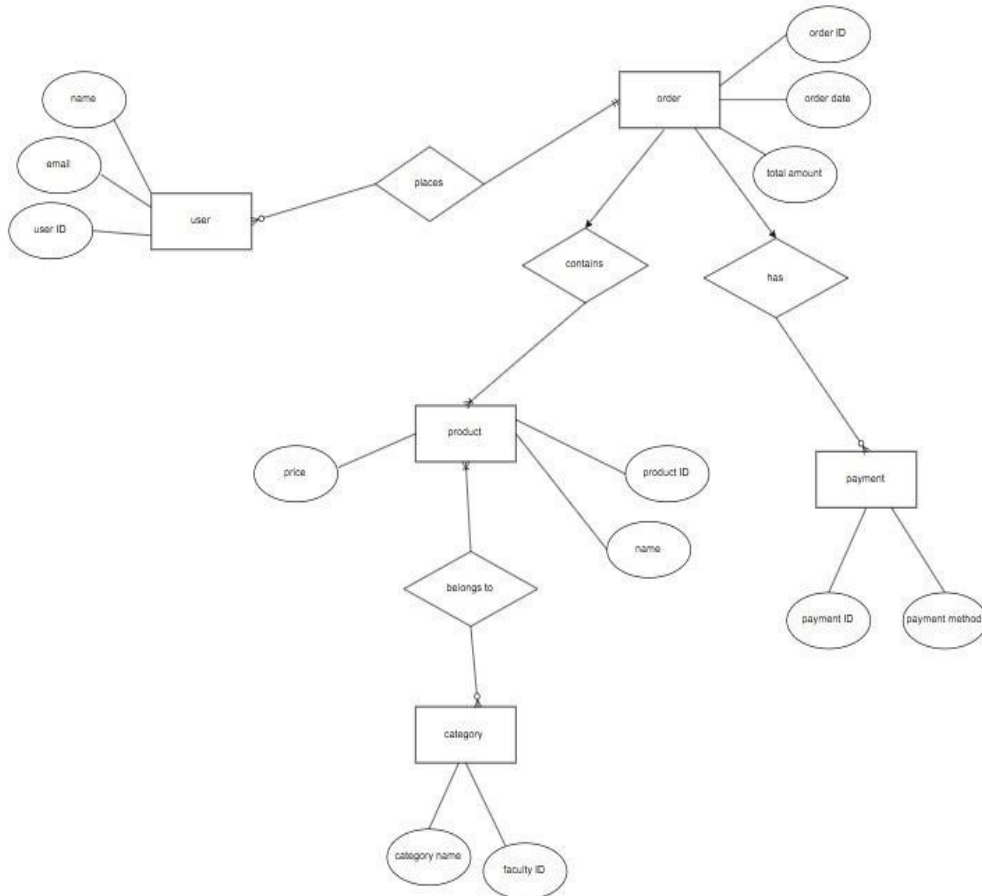
Breaks the main process into smaller parts like browsing products, order processing, and account management.

It shows internal data flows and connections between functions.



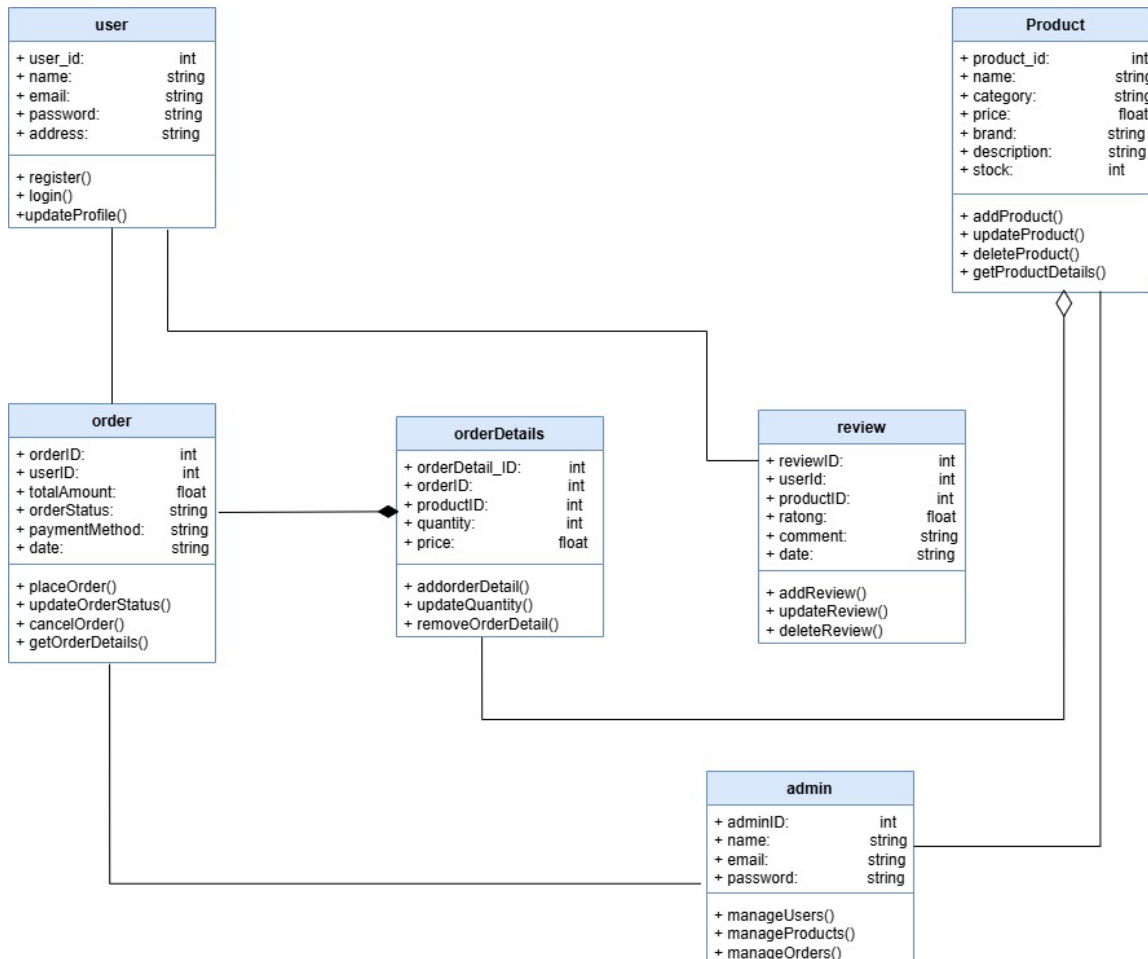
3. ER Diagram (Entity-Relationship Diagram)

The ER Diagram illustrates the relationships among system entities—Users, Products, Orders, Payments—depicting how data is structured and interconnected to support efficient data retrieval and operations like order processing and inventory management.



4. Class Diagram

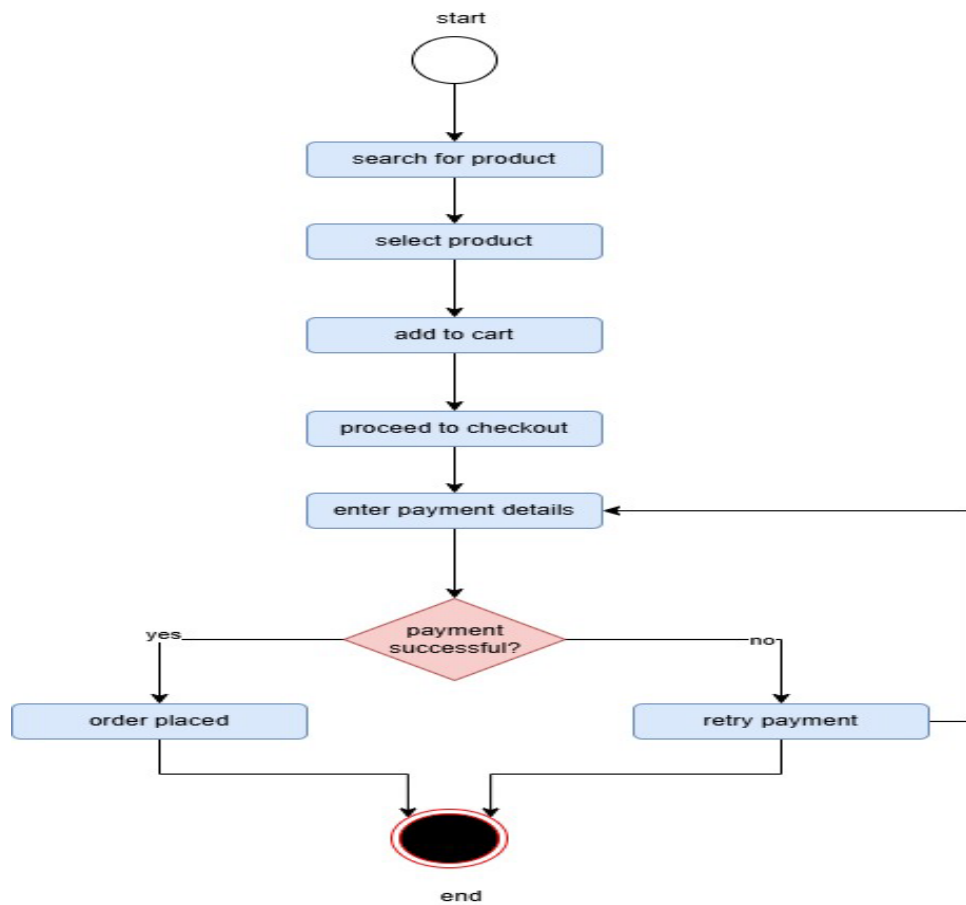
The Class Diagram defines the system's core entities, including User, Product, Cart, and Order. It illustrates relationships such as user orders and product inventory, and highlights attributes and methods associated with each class.



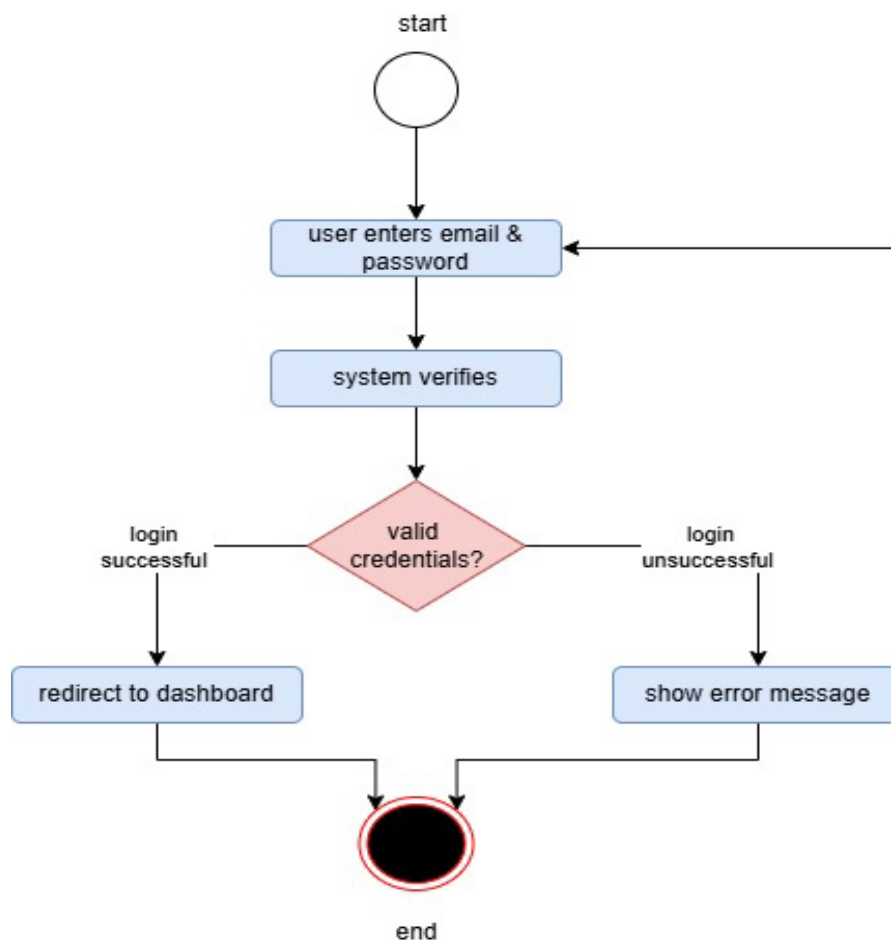
5. Activity Diagrams

These diagrams show how users move through the shopping process:

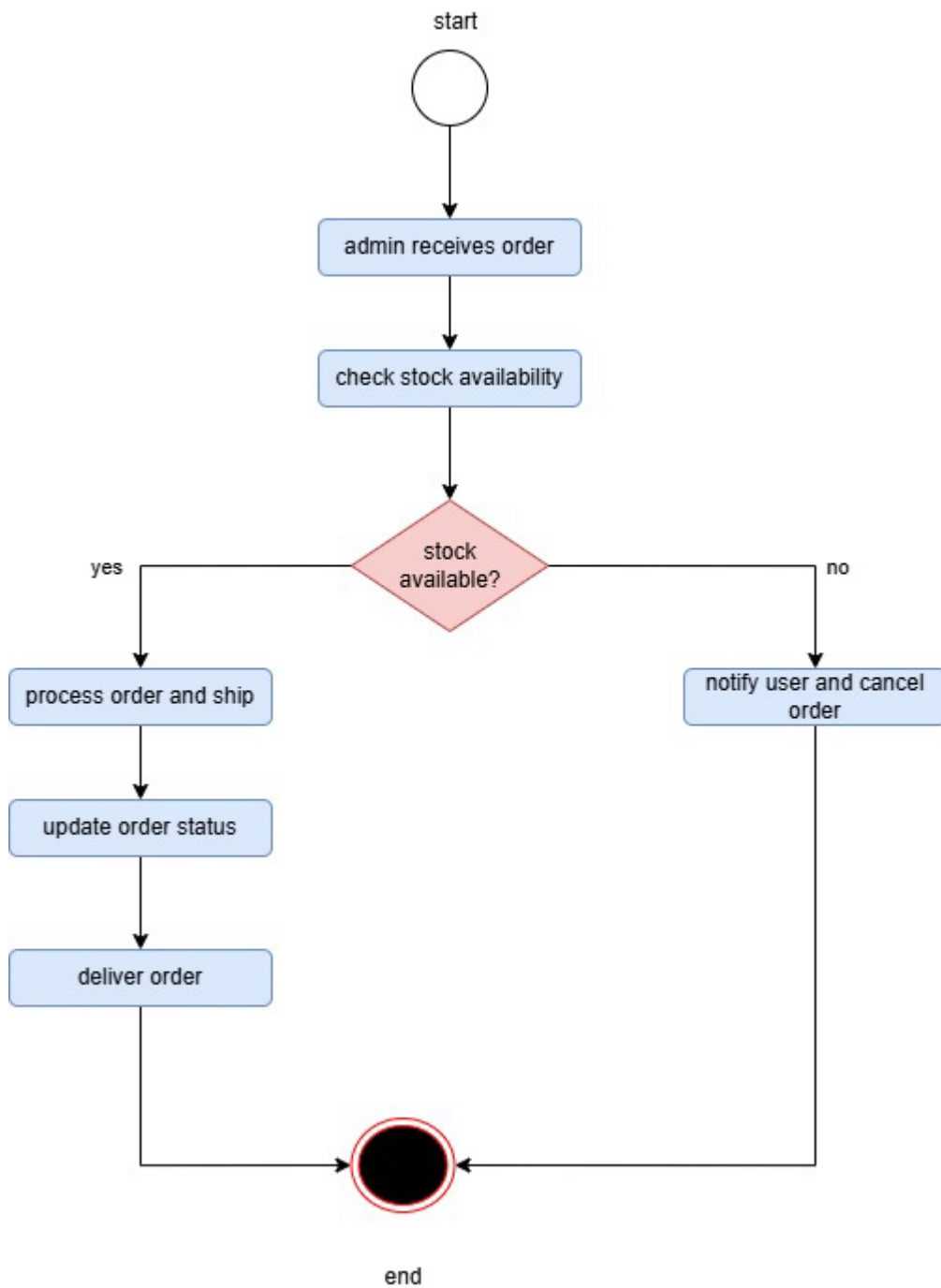
- **Activity 1: User Login** — User enters credentials to access the system.



- **Activity 2: Browsing and Purchasing Products** — User searches, views, and selects products to buy.



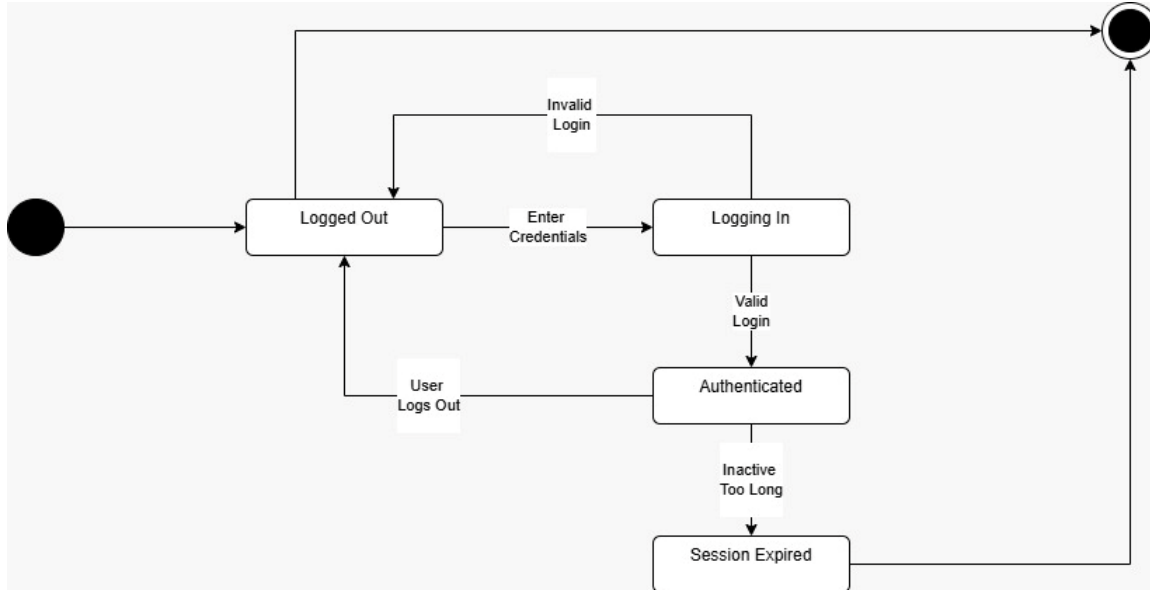
- **Activity 3: Order Processing and Management** — User completes payment; admin manages order status.



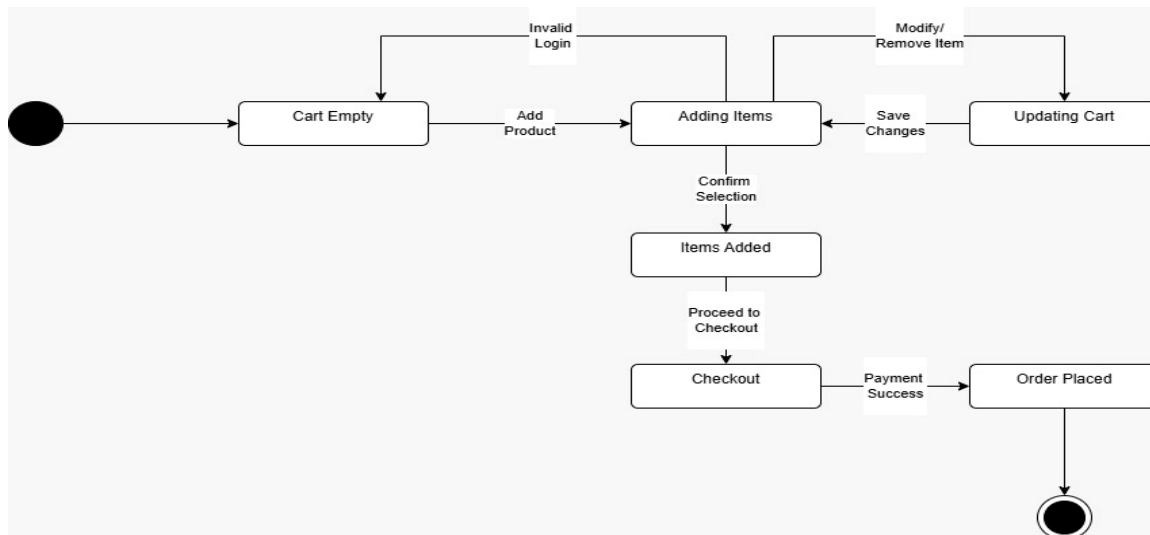
6. State Diagrams

Shows the main states during shopping:

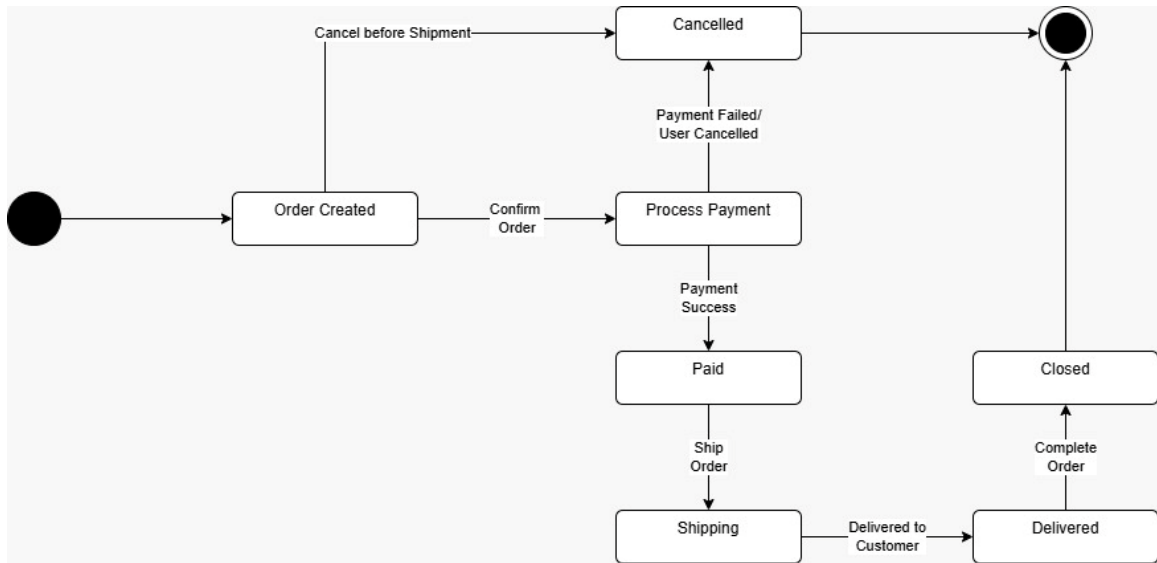
- **User Login State:** User logs into the system.



- **Shopping Cart State:** User adds items to the cart.

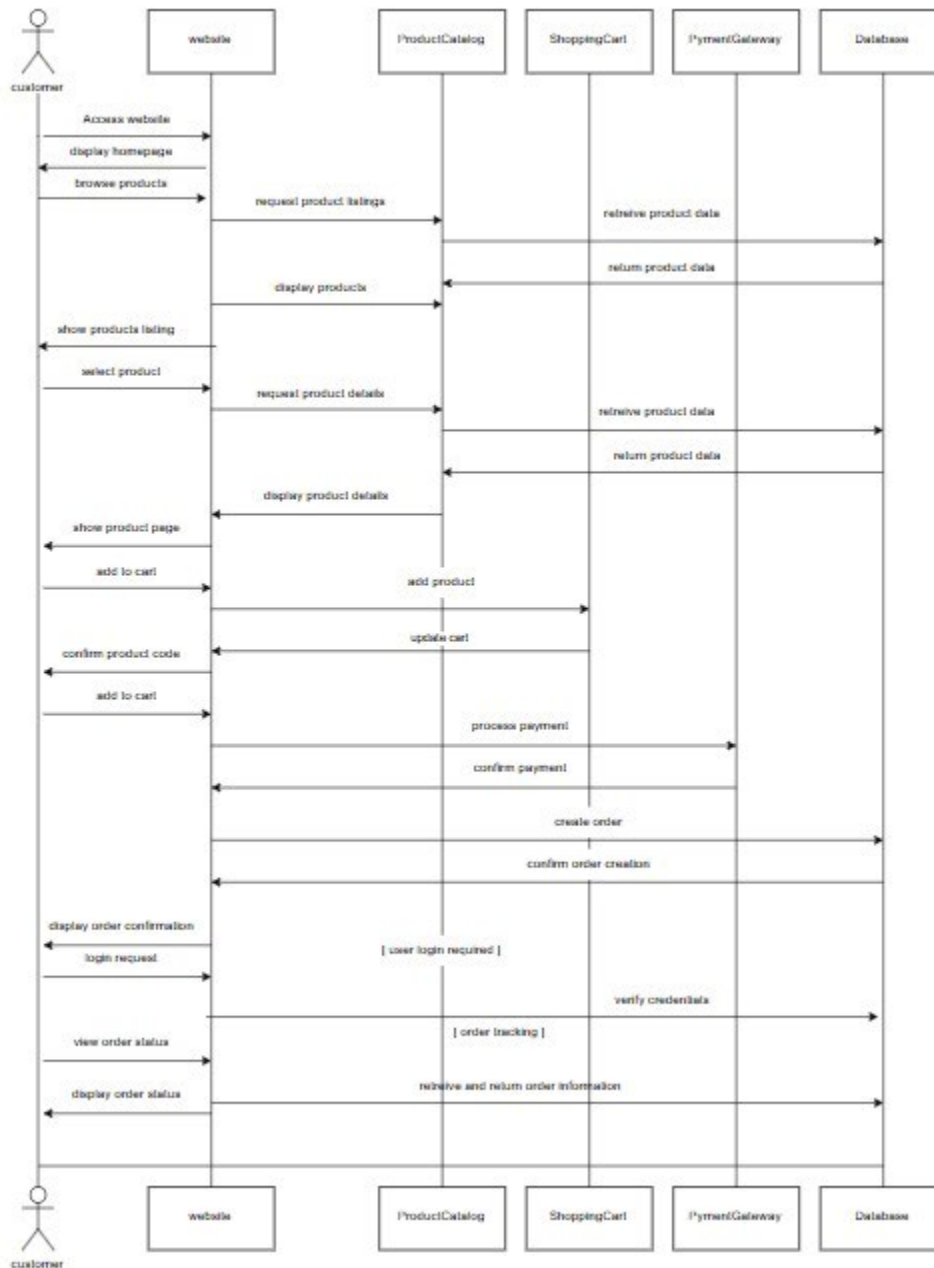


- **Order Process State:** User completes payment and places an order.



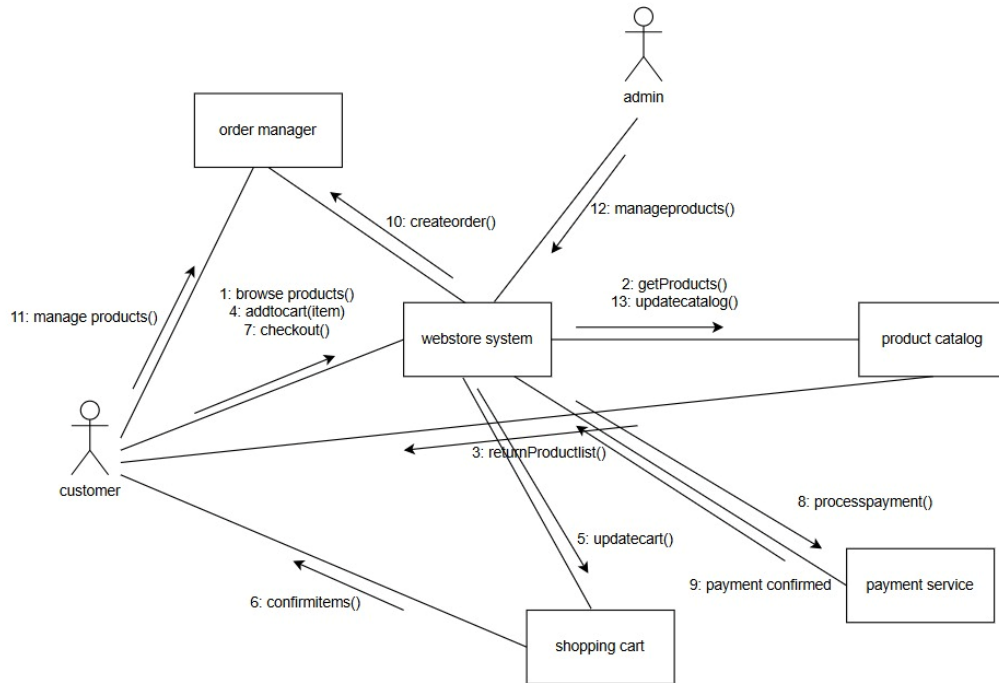
7. Sequence Diagrams

A sequence diagram shows how objects interact in a particular scenario of the system. It details the sequence of messages exchanged between system entities over time.



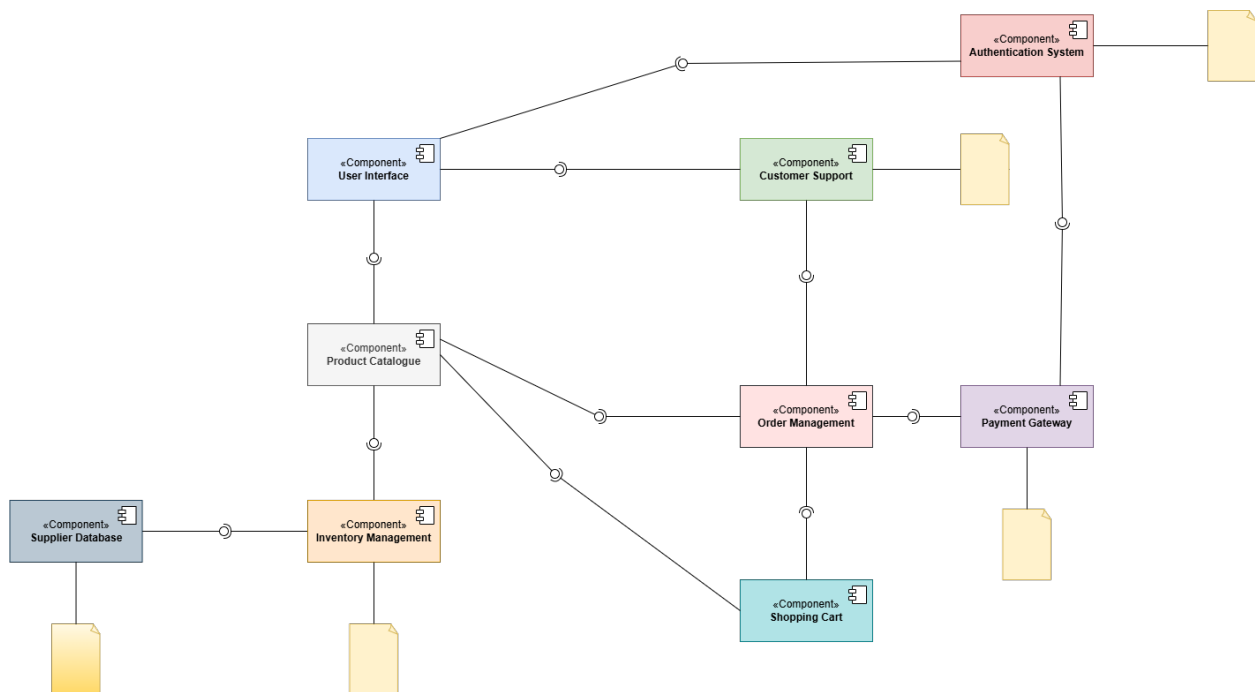
8. Collaboration Diagram:

A collaboration diagram represents interactions among objects, focusing on the relationships between them. It emphasizes how roles are linked and how messages are passed between them



9. Component Diagram:

The Component Diagram shows the main parts of the system and how they are connected. It helps understand how the software is divided into modules.



Design Considerations

1. Assumptions and Dependencies:

- Users have stable internet access.
- Payment services like PayPal/Visa are operational.
- Wix Studio will remain the development platform.

2. Limitations:

- Limited backend access due to Wix restrictions.
- Real-time inventory sync with third-party tools is constrained.

3. Design Constraints:

- Wix-specific plugins and structure must be used.
- System must remain user-friendly and lightweight.

Evaluation of the Design Phase

The design phase is complete and all required parts are included. The diagrams and descriptions are clear and well-made.

Final Decision: Approved with Minor Modifications

Only small improvements like adding wireframes or error handling can be done later. Overall, the work is good and ready to move forward.

Conclusion:

The Fiorella Beauty website successfully delivers a secure, fast, and easy-to-manage online store. It supports key e-commerce functions such as product display, checkout, and payment processing. The use of Wix Studio makes development simple and scalable, meeting both technical and business goals.